DIPLOMA PROGRAMME PRACTICE · SL · SET A

Chemistry / Paper 1B

Question booklet · 25 marks · 90 minutes across 1A and 1B

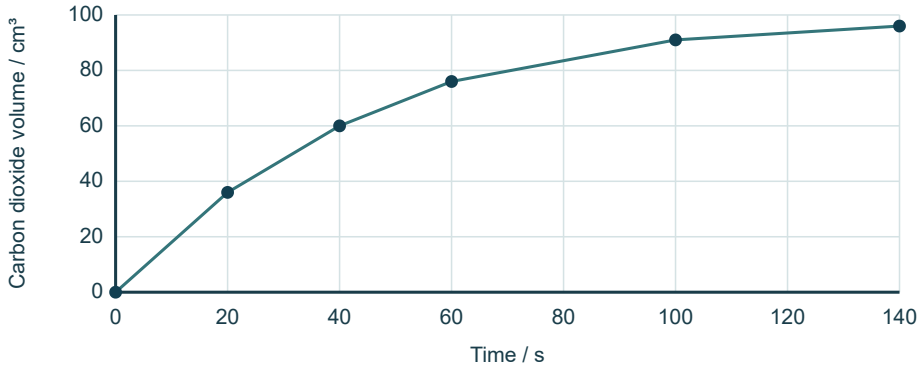
Answer every part. Show working, units and suitable precision. Use a calculator and a clean Chemistry data booklet. Use separate answer paper for additional working and diagrams.

QUESTION 1 [12]

Gas collection from a mineral

A 0.500 g powdered sample contains $CaCO_3$ and inert silicate. Excess hydrochloric acid is injected into an airtight flask connected to a gas syringe. At the measurement conditions the molar gas volume is 24.0 dm³ mol⁻¹. Use M($CaCO_3$)=100.1 g mol⁻¹. The points are cumulative CO_2 volumes; the final value is constant.

Time / s	0	20	40	60	100	140
CO_2 / cm³	0	36	60	76	91	96



Carbon dioxide volume / cm³ against Time / s. Data points are shown by filled circles.

(a) Write the net ionic equation and determine the mean rate of CO_2 production between 20 and 60 s. **[3]**

(b) Calculate the mass percentage of calcium carbonate in the sample. **[3]**

(c) The final collected volume has absolute uncertainty ±1 cm³ and the sample mass ±0.002 g. Treat other data as exact. Estimate the maximum absolute uncertainty in the percentage purity. **[3]**

Paper 1B · Question 1 (continued)

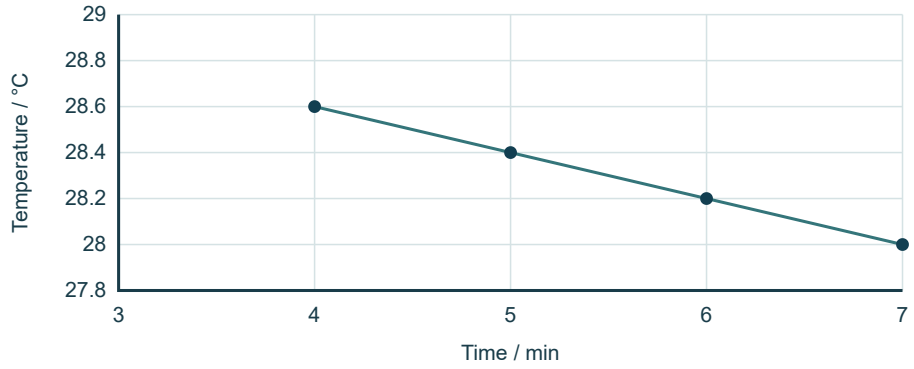
(d) A repeat with smaller particles reaches the same plateau sooner. Explain both observations and predict the effect of an unnoticed gas leak on the calculated purity. **[3]**

QUESTION 2 [13]

Recovering the mixing temperature

An insulated cup holds 50.0 cm³ of 1.20 mol dm⁻³ HCl. At t=3.0 min the student adds 50.0 cm³ of 1.20 mol dm⁻³ NaOH. Both solutions began at 21.0 °C. The cup has heat capacity 36.0 J K⁻¹. Use density 1.00 g cm⁻³ and specific heat capacity 4.18 J g⁻¹ K⁻¹ for the mixture. The post-mixing points follow a straight cooling line.

Time / min	4.0	5.0	6.0	7.0
Temperature / °C	28.6	28.4	28.2	28.0



Temperature / °C against Time / min. Data points are shown by filled circles.

(a) Use the cooling data to estimate the temperature immediately after mixing and the corrected temperature rise. **[3]**

(b) Calculate the energy gained by the solution and by the cup, separately. State which quantity must be used as the magnitude of the heat released. **[3]**

(c) Write the net ionic reaction and calculate ΔH per mole of water formed. **[3]**

(d) Explain the direction of bias if the first post-mixing temperature were used without extrapolation and the cup heat capacity were also ignored. **[2]**

Paper 1B · Question 2 (continued)

(e) Give one condition needed for this extrapolation and one experimental check of that condition. **[2]**